

Detailed Derivation of the Laplace Equation on a Disk Using Polar Coordinates

1 Problem Setup

Consider the two-dimensional Laplace equation

$$u_{xx} + u_{yy} = 0.$$

Polar coordinates

$$x = r \cos \theta, \quad y = r \sin \theta,$$

where

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan\left(\frac{y}{x}\right).$$

So

$$u(x, y) = u(r, \theta).$$

Goal:

$$\boxed{u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0}$$

From

$$r = \sqrt{x^2 + y^2},$$

we obtain

$$r_x = \frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r}.$$

Since

$$x = r \cos \theta,$$

we have

$$\boxed{r_x = \cos \theta}.$$

Similarly,

$$r_y = \frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}} = \frac{y}{r} = \sin \theta.$$

Thus

$$\boxed{r_y = \sin \theta}.$$

Since

$$\theta = \arctan\left(\frac{y}{x}\right),$$

we get

$$\theta_x = \frac{-y/x^2}{1 + y^2/x^2} = -\frac{y}{x^2 + y^2}.$$

Because

$$x^2 + y^2 = r^2$$

and

$$y = r \sin \theta,$$

we obtain

$$\theta_x = -\frac{\sin \theta}{r}.$$

Similarly,

$$\theta_y = \frac{1/x}{1 + y^2/x^2} = \frac{x}{x^2 + y^2}.$$

Using

$$x = r \cos \theta,$$

we obtain

$$\theta_y = \frac{\cos \theta}{r}.$$

Therefore, the four fundamental derivatives are

$$r_x = \cos \theta, \quad r_y = \sin \theta, \quad \theta_x = -\frac{\sin \theta}{r}, \quad \theta_y = \frac{\cos \theta}{r}.$$

2 Derive the First-Order Differential Operators

Let

$$u(x, y) = u(r(x, y), \theta(x, y)).$$

By the chain rule,

$$u_x = u_r r_x + u_\theta \theta_x.$$

Substituting the formulas above gives

$$u_x = \cos \theta u_r - \frac{\sin \theta}{r} u_\theta.$$

Thus

$$u_x = \cos \theta u_r - \frac{\sin \theta}{r} u_\theta.$$

Similarly,

$$u_y = u_r r_y + u_\theta \theta_y,$$

so

$$u_y = \sin \theta u_r + \frac{\cos \theta}{r} u_\theta.$$

Hence

$$u_y = \sin \theta u_r + \frac{\cos \theta}{r} u_\theta.$$

Equivalently, the **differential operators** themselves are

$$\boxed{\frac{\partial}{\partial x} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}}$$

and

$$\boxed{\frac{\partial}{\partial y} = \sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta}}.$$

3 Derive u_{xx} and u_{yy}

Starting with

$$u_x = \cos \theta u_r - \frac{\sin \theta}{r} u_\theta,$$

we apply

$$\frac{\partial}{\partial x} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}.$$

Thus

$$u_{xx} = \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\cos \theta u_r - \frac{\sin \theta}{r} u_\theta \right).$$

Hence

$$\boxed{\begin{aligned} u_{xx} = & \cos^2 \theta u_{rr} + \frac{\sin^2 \theta}{r} u_r - \frac{2 \sin \theta \cos \theta}{r} u_{r\theta} \\ & + \frac{2 \sin \theta \cos \theta}{r^2} u_\theta + \frac{\sin^2 \theta}{r^2} u_{\theta\theta}. \end{aligned}}$$

Similarly,

$$u_{yy} = \left(\sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\sin \theta u_r + \frac{\cos \theta}{r} u_\theta \right).$$

Therefore,

$$\boxed{\begin{aligned} u_{yy} = & \sin^2 \theta u_{rr} + \frac{\cos^2 \theta}{r} u_r + \frac{2 \sin \theta \cos \theta}{r} u_{r\theta} \\ & - \frac{2 \sin \theta \cos \theta}{r^2} u_\theta + \frac{\cos^2 \theta}{r^2} u_{\theta\theta}. \end{aligned}}$$

4 Laplace Equation in Polar Coordinates

Adding the two expressions gives

$$\begin{aligned} u_{xx} + u_{yy} = & (\cos^2 \theta + \sin^2 \theta) u_{rr} \\ & + \frac{\sin^2 \theta + \cos^2 \theta}{r} u_r \\ & + \left(-\frac{2 \sin \theta \cos \theta}{r} + \frac{2 \sin \theta \cos \theta}{r} \right) u_{r\theta} \\ & + \left(\frac{2 \sin \theta \cos \theta}{r^2} - \frac{2 \sin \theta \cos \theta}{r^2} \right) u_\theta \\ & + \frac{\sin^2 \theta + \cos^2 \theta}{r^2} u_{\theta\theta}, \end{aligned}$$

or

$$u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}.$$

Therefore,

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0.$$

An equivalent divergence-form expression is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0.$$